3. ANALYSIS OF STATICALLY INDETERMINATE STRUCTURES

General:

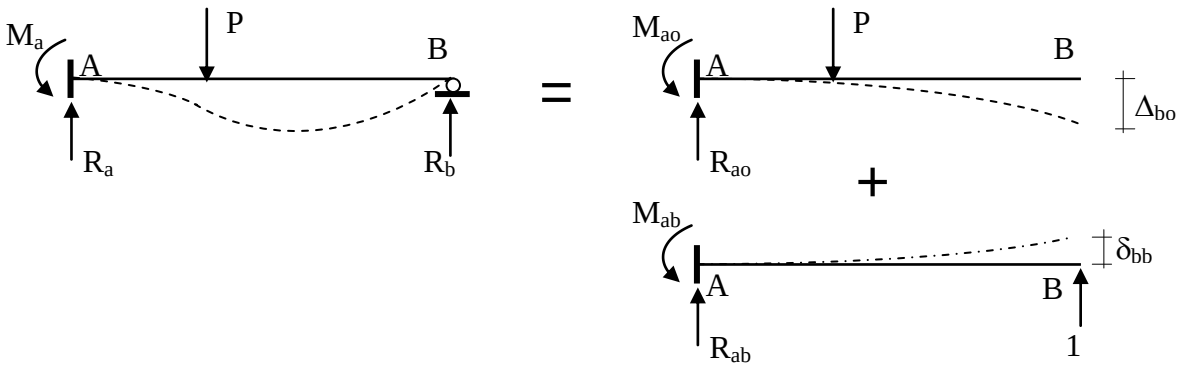
Any structure or framework systems in which the support reactions or internal forces in members can not be determined by using only the equations of static equilibrium are called ***Statically Indeterminate Structures***. Most real engineering structures are constructed in such a way that they are highly statically indeterminate for better rigidity and economy under service loads. Thus knowledge of analysis of indeterminate structures has become increasingly important as the use of these types of construction has become more extensive. There are various methods for analysis of statically indeterminate structures and some of them are discussed subsequently.

Consistent Deformation (superposition):

This method is based on satisfying compatibility of rotation and displacement at some points (preferably at redundant supports where the displacement and rotations can be known by inspection from the support mechanism) of the structure and it is suited to simple indeterminate structures, such as continuous beam. The redundant supports are identified and removed leaving only just sufficient restraints for stable equilibrium condition. The remaining statically determinate system is called the primary or base structure. Then the actual load and the unknown magnitudes of reaction forces at redundant supports are considered and the corresponding displacement/rotations are computed independently. Finally compatibility of rotation or displacement relation corresponding to each redundant supports is considered to determine the unknown reaction forces of redundant supports. In doing so, the rotation or displacement components of the base structure due to the applied load and the unknown reaction forces corresponding to the redundant supports are vectorially added and the resultant rotation or displacement vectors are equated to zero for unyielding support. Then the reactions at the remaining supports can be determined from equilibrium condition of the actual structure. The analysis of internal forces can be preceded with the usual procedure once

all reaction forces are determined. Such procedure of analyzing statically indeterminate structures by considering vectorial superposition of displacement components is also called method of superposition.

The procedure can best be elaborated and well understood by considering a singly redundant beam (propped cantilever) shown below:



The beam is statically indeterminate to the 1st degree. One of the three reaction components can be considered as being extra or redundant and let R_b be the redundant reaction. Suppose that we remove the vertical support at b and the vertical reaction R_b from the actual structure, the remaining statically determinate and stable cantilever beam is called the primary or base structure. The base structure can be considered to be subjected to the combined effect of the same loading as the actual structure plus the unknown redundant reaction force R_b . With this it can be seen that the condition of stresses in the actual structure and the primary structure are the same. Then the condition of deformation of the structure must also be exactly the same. Therefore the rotation and deflection at each point are identical for the two cases.

For example considering point b of the above beam:

Let Δ_b = deflection of point b of the actual structure

Δ_{bo} = deflection of point b of the primary structure due to the applied loads.

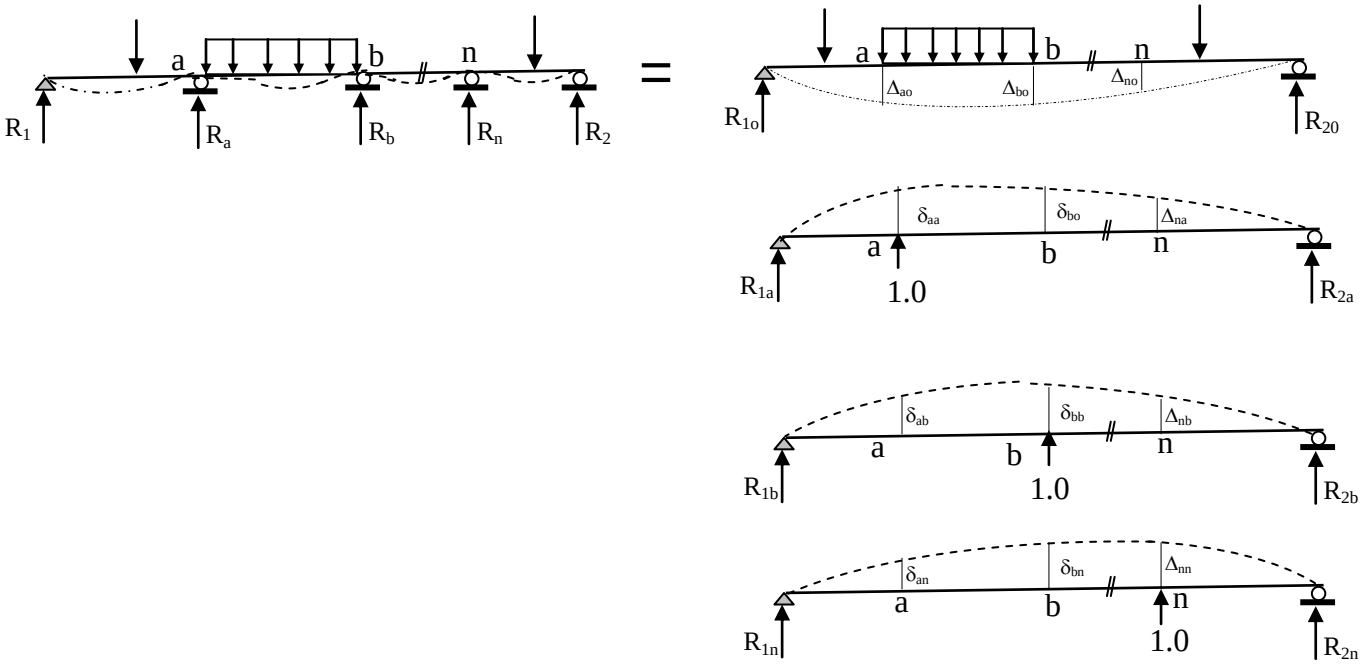
$\Delta_{bb} = R_b\delta_{bb}$ = deflection of point b of the base structure due to only the unknown magnitude redundant reaction.

Considering the algebraic sum of deflection at support:

$\Delta_b = \Delta_{bo} + \Delta_{bb} = \Delta_{bo} + R_b \delta_{bb}$, $\Delta_b = 0$ for un yielding support

$R_b = - \Delta_{bo} / \delta_{bb}$

The Application of consistent deformation /superposition principle for analysis of highly statically indeterminate structures can be generalized by considering the following continuous beam.



* The displacement at any point can be obtained by considering the combined effect of the n-loading cases corresponding to the reaction of the n-redundant supports and the base structure (principle of superposition).

$$\Delta_a = \Delta_{ao} + R_a \delta_{aa} + R_b \delta_{ab} + \ldots + R_n \delta_{an} = 0$$

$$\Delta_b = \Delta_{bo} + R_a \delta_{ba} + R_b \delta_{bb} + \ldots + R_n \delta_{bn} = 0$$

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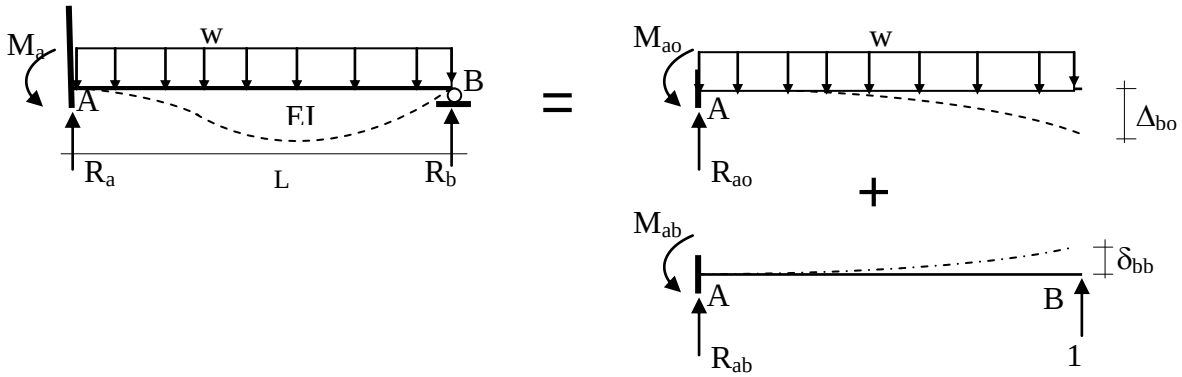
$$\Delta_n = \Delta_{no} + R_a \delta_{na} + R_b \delta_{nb} + \ldots + R_n \delta_{nn} = 0$$

The system of linear equation can be solved simultaneously (by appropriate method of solution) and the reaction at the redundant supports can be determined accordingly.

This general procedure can be applied to trusses with redundant supports or additional members beyond the minimum requirement for stable arrangement and rigid frames with redundant supports.

Example 1

Determine the reactions at the supports for the following beam.

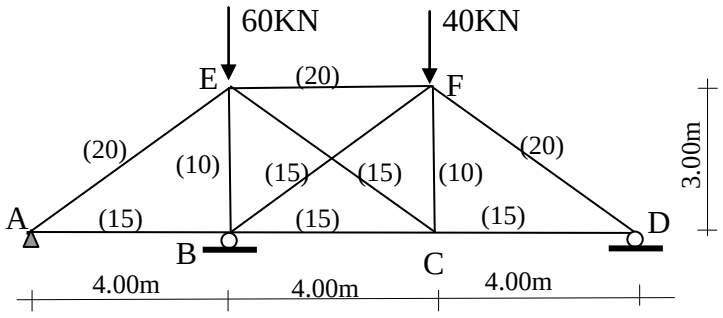


$$\Lambda_{bo} = -\frac{wL^4}{8EI} \; ; \quad \delta_{bb} = \frac{L^3}{3EI} \quad \Rightarrow \quad R_b = -\frac{\Lambda_{bo}}{\delta_{bb}} = -\frac{-wl^4/8EI}{L^3/3EI} = \frac{3}{8}wL$$

$$R_a = wL - \frac{3}{8}wL = \frac{5}{8}wL \quad ; \quad M_a = \frac{WL^2}{2} - \frac{3}{8}wL*L = \frac{wL^2}{8}$$

Example 2:

Determine the reaction at support B and the force in member BF for the following truss. The crossectional area of each member is given in bracket in cm2. assume E 210 Gpa for all members.



Soln.

- ***the truss is statically indeterminate to the 1st degree both externally and internally and hence to the 2nd degree in the overall condition***
- ***Consider the reaction at support B as redundant support and member BF redundant member to be removed so that the remaining structure will be statically determinate both externally and internally and serve as the base structure***
- ***Analyze the base structure due to the real loads and determine member forces.***
  - ***Apply unit load in the direction of reaction force at support B and determine member forces due to the unit load.***
  - ***Apply pair of unit loads in the direction of member BF and determine member forces due to these unit loads.***
  - ***Compute displacements of the base structure in the vertical direction at joint b and in the direction of member BF due to the actual loads, unit load at joint B in the vertical direction and pair of unit loads in the direction of member BF.***

-Formulate displacement compatibility relation at joint B in the vertical direction and along member BF and solve for the unknown force at redundant support and in the redundant member BF.

The computation can be done much more conveniently in tabular form as follows:

Member	L (m)	A (cm^2)	N (KN)	n_B	n_{BF}	$n_B \frac{NL}{A}$	$n_{BF} \frac{NL}{A}$	$\frac{n_B^2 L}{A}$	$\frac{n_B n_{BF} L}{A}$	$\frac{n_{BF}^2 L}{A}$
AE	5.0	20	-88.89	+1.12	0.00	-24.89	0.00	+0.31	0.00	0.00
EF	4.0	20	-62.20	+0.45	-0.80	-5.60	+9.95	+0.04	-0.07	+0.13
DF	5.0	20	-77.75	+0.56	0.00	-10.89	0.00	+0.08	0.00	0.00
DC	4.0	15	+62.20	-0.45	0.00	7.46	0.00	+0.05	0.00	0.00
BC	4.0	15	+71.10	-0.89	-0.80	-16.87	-15.17	+0.21	+0.19	+0.17
AB	4.0	15	+71.10	-0.89	0.00	-16.87	0.00	+0.21	0.00	0.00
BE	3.0	10	0.00	-1.00	-0.60	0.00	0.00	+0.30	+0.18	+0.11
BF	5.0	15	0.00	0.00	+1.00	0.00	0.00	0.00	0.00	+0.33
EC	5.0	15	-11.10	+0.55	+1.00	-2.04	-3.7	+0.10	+0.18	+0.33
CF	3.0	15	-6.65	-0.33	-0.60	+0.60	+1.20	+0.03	+0.06	+0.11
					Σ	**-83.94**	**-7.72**	**+1.33**	**+0.54**	**+1.18**

The compatibility condition for vertical displacement of joint B and displacement in the direction of member BF of the truss are stated as follows:

$$\Delta_B = \Delta_{bo} + R_B \delta_{bb} + F_{BF} \delta_{b,BF} = 0$$

$$\Delta_{BF} = \Delta_{BF,o} + R_B \delta_{b,BF} + F_{BF} \delta_{BF,BF} = 0$$

$$-83.94 + 1.33R_B + 0.54F_{BF} = 0$$

$$-7.72 + 0.54R_B + 1.18F_{BF} = 0$$

Solving the above simultaneous equations:

R_B = 74.24KN (upward)

F_{BF} = - 27.40KN (compression)